

4 ZFS file system

A file system is responsible for managing files and operations that involve files. But even though most of the things we do on our computer involves a file system either directly or indirectly, most of us hardly pay attention to the file systems of our operating system. In this section, we will explore one of the most interesting aspects of OpenSolaris, which is its file system, the ZFS file system. Tools like Time Slider and Boot Environments make use of the capabilities ZFS file systems has, in order to provide the unmatched features in OpenSolaris. In the following sections, we will explore the ZFS file system, its features, usage, etc. by presenting relevant portions from the Solaris ZFS Administration Guide, published by Sun Microsystems Inc. This guide is also available on Digit DVD or you can download it from <http://tr.im/zfsguide>.

For trying out exercises in this chapter, we recommend you assume root user role. This can be done by executing `$ pfexec su -` in a terminal.

If you do not understand any term related to ZFS, then look for in the ZFS Terminology table later in the chapter.

4.1 What is ZFS?

The ZFS file system is a revolutionary new file system that fundamentally changes the way file systems are administered, with features and benefits not found in any other file system available today. ZFS has been designed to be robust, scalable, and simple to administer.

4.2 ZFS pooled storage

ZFS uses the concept of storage pools to manage physical storage. Historically, file systems were constructed on top of a single physical device. To address multiple devices and provide for data redundancy, the concept of a volume manager was introduced to provide the image of a single device so that file systems would not have to be modified to take advantage of multiple devices. This design added another layer of complexity and ultimately prevented certain file system advances, because the file system had no control over the physical placement of data on the virtualised volumes.

ZFS eliminates the volume management altogether. Instead of forcing you to create virtualised volumes, ZFS aggregates devices into a storage pool. The storage pool describes the physical characteristics of the storage (device